asar: :CHEATSHEET

Partially automate a U.S. fisheries stock assessment report.

Install

Install 4 R packages: pak, tinytex, asar, and stockplotr

```
install.packages("pak")
pak::pak("tinytex")
tinytex::install_tinytex(bundle = "TinyTeX-2")
pak::pak("nmfs-ost/asar")
pak::pak("nmfs-ost/stockplotr")
```

Ensure Quarto CLI v1.6+ is installed

```
quarto::quarto_version()
```

Create report skeleton

Make report with minimal recommended arguments

```
asar::create_template(
  office = "NWFSC",
  region = "U.S. West Coast",
  species = "Petrale sole",
  spp_latin = "Eopsetta jordani",
  year = 2023,  # specify if past year
  authors = c("John Snow" = "NWFSC"),
  model_results = here::here(" model_output.rda")
)
```

Model output file converted with `stockplotr::convert_output()`

report

- 01_executive_summary.qmd
- 02_introduction.qmd
- ...
- 11_appendix.qmd

child .qmd files

- asar_references.bib — *reference list*
- preamble.R — *key quantities to reference in report*
- model_output_metadata.md — *records location & creation date of `model_output.rda`*
- report_glossary.tex — *list of acronyms & meanings, including survey names*
- SAR_USWC_Petrale_sole_skeleton.qmd — *file rendered into report*
- support_files/ — *files used for cover page & structure*

Use time-saving shortcuts

Save time writing reports by using features including quantities, acronyms, citations, survey names, and parameters.

03_data.qmd

```
# Data {#sec-data}

For `r end_year`, composition data was
obtained from the \gls{pacfin}
Biological Data System
[@star_1999].

During the \gls{ny dec nearshore},
`r species` was observed...
```

End year calculated in **preamble.R**

Acronym in **report_glossary.tex**

Citation in **asar_references.bib**

Informal survey name, used like acronyms in **report_glossary.tex**, will render as correct survey name

Parameter in **report skeleton yaml** (e.g., **SAR_USWC_Petrale_sole_skeleton.qmd**)

Add tables & figures

Steps:

1. Create tables, figures with `stockplotr` or other method
2. Place table .rda files in "tables" folder
3. Place figure .png, .jpg, or .rda files in "figures" folder
4. Add code chunks to `report/08_tables.qmd` that import and display tables in "tables" folder:

```
asar::create_tables_doc(
  subdir = here::here("report"),
  tables_dir = here::here()
)
```

"tables" folder location

5. Update `09_figures.qmd` with similar workflow:

```
asar::create_figures_doc(
  subdir = here::here("report"),
  figures_dir = here::here()
)
```

Location to save `09_figures.qmd`

Render report

Options:

- Use R console

```
quarto::quarto_render(
  here::here("report",
  "SAR_USWC_Petrale_sole_skeleton.qmd")
)
```

Report skeleton

- RStudio "render" button in program pane

- Use terminal: `quarto render report/skeleton.qmd`

Add accessibility features

For PDFs: Add tags & alternative text for figures

```
add_accessibility(
    x = "SAR_USWC_Petrale_sole_skeleton.tex",
    dir = getwd(),
    alttext_csv = file.path(getwd(),
"captions_alt_text.csv"),
    compile = TRUE
  )
```

LaTeX version of report created by rendering skeleton

File with templates for captions and alternative text. Templates must be reviewed and expanded by user

Captions/alternative text file format *(captions_alt_text.csv)*

label	type	caption	alt_text
example_fig	figure	Example caption for figure.	Example alternative text for figure.
example_tab	table	Example caption for table.	
...	...	...	...

See "Increasing Report Accessibility" vignette

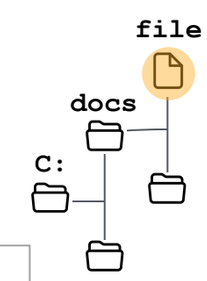
Tip

Use relative filepaths

- Saves time and effort
- Facilitates collaboration

Recommended tool: `here` R package

```
absolute_path <- "C:/docs/file"
relative_path <- here::here("file")
```



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